

```
use joinDB;
CREATE TABLE departments (
    dept_id INT AUTO_INCREMENT,
    dept_name VARCHAR(100) NOT NULL,
    PRIMARY KEY (dept_id)
) AUTO_INCREMENT = 101;

CREATE TABLE employees (
    emp_id INT AUTO_INCREMENT,
    emp_name VARCHAR(100) NOT NULL,
    salary DECIMAL(10,2),
    dept_id INT,
    PRIMARY KEY (emp_id),
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
) AUTO_INCREMENT = 10001;
```

```
INSERT INTO departments (dept_name) VALUES
('Human Resources'),
('Finance'),
('IT'),
('Marketing'),
('Sales');
```

```
INSERT INTO employees (emp_name, salary, dept_id) VALUES
('Amit Sharma', 55000, 101),
('Neha Verma', 60000, 101),
('Rohit Mehta', 75000, 102),
('Pooja Singh', 72000, 102),
('Karan Patel', 80000, 103),
('Anjali Kapoor', 82000, 103),
('Vikas Gupta', 68000, 104),
('Sneha Reddy', 70000, 104),
('Rahul Nair', 90000, 105),
('Priya Iyer', 88000, 105),
('Arjun Das', 62000, 101),
('Kavita Joshi', 64000, 102),
('Manish Yadav', 76000, 103),
('Ritika Jain', 71000, 104),
('Deepak Choudhary', 83000, 105),
('Simran Kaur', 59000, 101),
('Varun Malhotra', 67000, 102),
('Nikita Bansal', 79000, 103),
('Suresh Pillai', 72000, 104),
('Ayesha Khan', 91000, 105);
```

```
select * from employees;
select * from departments;
```

```

create table employees_manager(emp_id int primary key auto_increment,
emp_name varchar(225), manager_id int,
foreign key(manager_id) references employees_manager(emp_id)
on delete set null on update cascade
);

```

```

INSERT INTO employees_manager (emp_name, manager_id) VALUES
('Rohit Sharma', NULL),      -- CEO
('Anita Verma', 1),          -- Manager
('Vikas Singh', 1),
('Priya Kapoor', 2),
('Rahul Mehta', 2),
('Sneha Joshi', 3),
('Karan Patel', 3),
('Amit Gupta', 4),
('Neha Reddy', 4),
('Pooja Nair', 5),
('Arjun Das', 5),
('Simran Kaur', 6),
('Varun Malhotra', 6),
('Ritika Jain', 7),
('Deepak Sharma', 7),
('Kavita Bansal', 8),
('Manish Yadav', 8),
('Nikita Sinha', 9),
('Suresh Pillai', 9),
('Ayesha Khan', 10);

```

-- Joins:-

-- A join is used to combine rows from two or more tables based on related column.
 -- usually the relation is in Primary key in one table and foreign key in another table

-- Types of joins:-

-- inner join, left outer join, right outer join, left outer null join, right outer null join
 -- full join(Not present in MySQL, can be simulated), cross join, self join , natural join

-- 1) inner join(simple join):- 15 percent
 -- Return only matching rows in both tables.
 -- Syntax :-

```

/*
select column1, column2,.. columnN
from table1 t1

```

```
join table2 t2
on t1.column1 = t2.coloumn2;
*/
```

```
select *
from employees
inner join departments
on employees.dept_id = departments.dept_id;
```

```
select employees.dept_id, emp_name, dept_name
from employees
inner join departments
on employees.dept_id = departments.dept_id;
```

```
select d.dept_id, emp_name, dept_name
from employees as e
inner join departments as d
on e.dept_id = d.dept_id;
```

```
select d.dept_id, emp_name, dept_name
from employees e
inner join departments d
on e.dept_id = d.dept_id;
```

```
select d.dept_id, emp_name, dept_name
from departments d
join employees e
on e.dept_id = d.dept_id;
```

```
-- 2) Left outer join(left join):- 80 percent
-- left join returns all the rows from the left talbe and
-- matching rows from right table.
-- If no match is found then the right side of the table is populated
-- with null
```

```
select * from employees;
select *
from departments d
left outer join employees e
on e.dept_id = d.dept_id;
```

```
select d.dept_id, d.dept_name, e.emp_name
from departments d
left join employees e
on e.dept_id = d.dept_id;
```

```
-- 3) Right join:-  
-- Opposite of left join  
-- Returns all rows from right table and matching rows from left table
```

```
-- Find all the employees and only show the departments they are  
-- working in.
```

```
select *  
from departments d  
right join employees e  
on d.dept_id = e.dept_id;
```

```
-- 4) Left null join:-
```

```
-- find all the departments where no employee works
```

```
select * from employees;  
select * from departments;
```

```
select d.dept_id, dept_name  
from departments d  
left join employees e  
on e.dept_id = d.dept_id  
where e.dept_id is null;
```

```
-- find all the employees who doesn't belong to any department
```

```
select emp_id, emp_name  
from employees e  
left join departments d  
on d.dept_id = e.dept_id  
where e.dept_id is null;
```

```
-- 5) Right null join:-
```

```
-- find all the employees who doesn't belong to any department
```

```
select *  
from departments d  
right join employees e  
on d.dept_id = e.dept_id  
where e.dept_id is null;
```

```
-- 6) Cross join:-
```

```
-- returns cartesian product.
```

```
-- Every row of table1 combines with every row of table2.
```

```
select *  
from employees  
cross join departments;
```

```
-- 7) Self join:-
```

```
select * from employees_manager;
```

```
select em1.emp_id as manger_id, em1.emp_name as manager_name,  
em2.emp_id as employee_id, em2.emp_name as employee_name  
from employees_manager em1  
join employees_manager em2  
on em1.emp_id = em2.manager_id;
```

```
-- create a table secret santa where  
-- one employee gives a gift to another employee  
-- and you need to find the name of the santa as well as the reciver name  
-- secret_santa_table  
-- emp_id          emp_name          santa_id
```

```
-- 8) Natural join:-
```

```
select * from employees;  
select * from departments;
```

```
select *  
from employees  
natural join departments;
```